Problem Set 2

Econometrics

Code: https://github.com/ae4292/Econometrics_PSets

Problem 1

(a) Regression of lwage on the 30 education × age dummies (898 observations). Estimates $\hat{\alpha}_{e,a}$ with standard errors in parentheses:

	$a = 26$	$a = 27$	$a = 28$	$a = 29$	$a = 30$
$e = 12$	9.451 (0.091)	9.300 (0.085)	9.538 (0.083)	9.386 (0.085)	9.628 (0.088)
$e = 13$	9.612 (0.192)	9.756 (0.192)	9.378 (0.175)	9.820 (0.224)	9.446 (0.203)
$e = 14$	9.581 (0.224)	9.794 (0.171)	9.642 (0.216)	9.712 (0.175)	9.645 (0.171)
$e = 15$	9.535 (0.242)	9.386 (0.232)	9.573 (0.265)	9.540 (0.252)	9.919 (0.242)
$e = 16$	10.070 (0.167)	9.944 (0.187)	9.992 (0.148)	10.179 (0.131)	10.071 (0.164)
$e = 17$	9.958 (0.483)	10.046 (0.224)	10.166 (0.296)	9.618 (0.418)	10.016 (0.374)

α is the individual effect of having a given age and education on log wage.

(b)

$$\begin{aligned}
&\alpha_{e,a-1} - 2\alpha_{e,a} + \alpha_{e,a+1} = 0, && e = 12,\dots,17,\ a = 27,28,29,\\
&(\alpha_{e-1,27} - \alpha_{e-1,26}) - 2(\alpha_{e,27} - \alpha_{e,26}) + (\alpha_{e+1,27} - \alpha_{e+1,26}) = 0, && e = 13,\dots,16,\\
&\alpha_{e-1,26} - 2\alpha_{e,26} + \alpha_{e+1,26} = 0, && e = 13,\dots,16,
\end{aligned}$$

which gives R of size 26×30 and $q = 0$.

> $F = 0.920, \qquad p\text{-value} = 0.581$

I would not reject the null hypothesis as the p-value (0.58) is not below 0.05 significance level.

Problem 2

(a) $\mathbb{E}(\epsilon_i \mid x_i) = 0$, so $\mathbb{E}(\epsilon_i x_i) = \mathbb{E}\big(x_i\,\mathbb{E}(\epsilon_i \mid x_i)\big) = 0$ and $\hat{\beta} - \beta = \dfrac{\sum x_i \epsilon_i}{\sum x_i^2}$ is conditionally unbiased. The sum of even numbers squared is

$$\sum_{i=1}^{n} x_i^2 = 0 + 2^2 + 0 + 4^2 + \cdots + n^2 = \sum_{k=1}^{n/2} (2k)^2 = \frac{n(n+1)(n+2)}{6} \;\to\; \infty$$

(for n even). Therefore

$$\operatorname{Var}(\hat{\beta} \mid x) = \frac{\sigma^2}{\sum x_i^2} = \frac{6}{n(n+1)(n+2)} \to 0,$$

so $\hat{\beta}$ is unbiased with variance going to zero: OLS $\hat{\beta}$ for β is consistent, $\mathbb{E}[\hat{\beta}] = \beta$.

(b)

$$\sum_{i=1}^{n} x_i^2 = \sum_{i=1}^{n} \lambda^{2i} = \frac{\lambda^2 (1 - \lambda^{2n})}{1 - \lambda^2} \;\to\; \frac{\lambda^2}{1 - \lambda^2} < \infty$$

and

$$\mathrm{Var}(\hat{\beta} \mid x) = \frac{\sigma^2}{\sum x_i^2} \;\to\; \frac{1 - \lambda^2}{\lambda^2} \neq 0,$$

this however never goes to zero as $n \to \infty$, thus $\hat{\beta}$ does not converge in probability to β as the variance stays positive.

(c) For 2a we can have perfect collinearity since for $n = 1$, $x_1 = 0$ and $\mathrm{rank}(X) \neq 1$. However for 2b Gauss-Markov holds. Both have strict exogeneity, homoskedasticity and no serial correlation.

(d) Since ε_i are normal, $\hat{\beta} - \beta = \dfrac{\sum x_i \varepsilon_i}{\sum x_i^2} \sim N\left(0, \dfrac{\sigma^2}{\sum x_i^2}\right)$ exactly, and $\sum x_i^2 \to \lambda^2/(1 - \lambda^2)$ from 2b, so with $\sigma^2 = 1$

$$\hat{\beta}_{OLS} \to N\left(\beta, \frac{1 - \lambda^2}{\lambda^2}\right).$$

> $\hat{\beta}_{OLS} \to N\left(\beta, \dfrac{1 - \lambda^2}{\lambda^2}\right)$

Problem 3

(a) $\theta = \dfrac{s_1^2}{\sigma^2} = 1, \quad g(\theta) = \log(\theta), \quad g'(\theta) = \dfrac{1}{\theta} = 1.$

$$\sqrt{n}(\hat{\theta} - \theta) \to N\big(0, (g'(\theta))^2 \cdot V\big) = N(0, 1 \cdot V) = N(0, 2)$$

$$\frac{s_1^2}{\sigma^2} = \frac{\chi^2_{n-1}}{n-1}, \qquad \mathrm{Var}\left(\sqrt{n}\,\frac{s_1^2}{\sigma^2}\right) = n\,\mathrm{Var}\left(\frac{\chi^2_{n-1}}{n-1}\right) = \frac{2n}{n-1} \xrightarrow[n\to\infty]{} 2$$

$$\begin{pmatrix} \sqrt{n_1}\log\dfrac{s_1^2}{\sigma^2} \\ \sqrt{n_2}\log\dfrac{s_2^2}{\sigma^2} \end{pmatrix} = N\left(\begin{pmatrix} 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix}\right) \qquad \text{independent samples no cov.}$$

$$\log(s_1^2) - \log(s_2^2) = \log(s_1^2) - \log(\sigma^2) - \log(s_2^2) + \log(\sigma^2) = \log\left(\frac{s_1^2}{\sigma^2}\right) - \log\left(\frac{s_2^2}{\sigma^2}\right)$$

$$T = \sqrt{\frac{n_2}{n_1 + n_2}} \cdot \sqrt{n_1}\log\left(\frac{s_1^2}{\sigma^2}\right) - \sqrt{\frac{n_1}{n_1 + n_2}}\,\sqrt{n_2}\log\left(\frac{s_2^2}{\sigma^2}\right)$$

$$\mathbb{E}(T) = \sqrt{\frac{n_2}{n_1 + n_2}} \cdot \mathbb{E}\left(\sqrt{n_1}\log\frac{s_1^2}{\sigma^2}\right) - \sqrt{\frac{n_1}{n_1 + n_2}}\,\mathbb{E}\left(\sqrt{n_2}\log\frac{s_2^2}{\sigma^2}\right) = 0$$

$$\mathrm{Var}(T) = \frac{n_2}{n_1 + n_2}\,\mathrm{Var}\left(\sqrt{n_1}\log\frac{s_1^2}{\sigma^2}\right) + \frac{n_1}{n_1 + n_2}\,\mathrm{Var}\left(\sqrt{n_2}\log\frac{s_2^2}{\sigma^2}\right) = 2 \cdot \frac{n_2 + n_1}{n_1 + n_2} = 2$$

(b) The asymptotic test rejects if $T > z_{0.95}\sqrt{2}$. The finite-sample test rejects if $\dfrac{s_1^2}{s_2^2} > q_{0.95}$, the 0.95 quantile of $F(n_1-1, n_2-1)$, that is

$$\sqrt{\frac{n_1 n_2}{n_1+n_2}}\,\log\frac{s_1^2}{s_2^2} > c_n := \sqrt{\frac{n_1 n_2}{n_1+n_2}}\,\log(q_{0.95}), \qquad \text{i.e. } T > c_n.$$

Under the null the F test has size exactly 0.05, so $P(T > c_n) = 0.05$ for every n. Also

$$F(n_1-1, n_2-1) = \frac{s_1^2/\sigma^2}{s_2^2/\sigma^2} \text{ as } n_1, n_2 \to \infty \;\to\; \frac{s_1^2}{s_2^2}\cdot c \sim \frac{s_1^2}{s_2^2}$$

so $T \to N(0,2)$, which has a continuous cdf, hence $c_n \to z_{0.95}\sqrt{2}$. Then

$$P(\text{tests disagree}) = P\big(T \text{ between } c_n \text{ and } z_{0.95}\sqrt{2}\big) \to 0,$$

that is $P(T > z_{0.95}\sqrt{2}) = 0.05$ is equivalent to $P(T > c_n) = 0.05$ as they have same probability as long as n_1 and $n_2 \to \infty$.

(c)

$$\mathbb{E}(T) = \sqrt{\frac{n_2}{n_1+n_2}}\,\mathbb{E}\left(\sqrt{n_1}\log\frac{s_1^2}{\sigma^2}\right) - \sqrt{\frac{n_1}{n_1+n_2}}\,\mathbb{E}\left(\sqrt{n_2}\log\frac{s_2^2}{\sigma^2}\right)$$

$$y_{ji} = \mu_j + \sigma z_{ji}, \qquad \mathbb{E}z = 0,\ \mathbb{E}z^2 = \int y^2\,dF_0(y) = 1,\ \mathbb{E}z^4 = \kappa$$

$$s_1^2 = \frac{1}{n_1-1}\sum\left(y_{1i}-\bar{y}_1\right)^2 = \frac{\sigma^2}{n_1-1}\sum(z_{1i}-\bar{z}_1)^2$$

$$\frac{s_1^2}{\sigma^2} = \frac{\sum z_i^2 - n_1\bar{z}^2}{n_1-1} = \frac{n_1}{n_1-1}\left(\frac{1}{n_1}\sum z_i^2 - \bar{z}^2\right)$$

As $n_1 \to \infty$, $\bar{z} = \dfrac{1}{n_1}\sum z_i \to 0$ and $\dfrac{1}{n_1}\sum z_i^2 \to 1$ as $\int y^2\,dF_0(y) = 1$, so

$$\frac{s_1^2}{\sigma^2} \xrightarrow{p} 1.$$

Write $m_2 = \frac{1}{n}\sum z_i^2$ (dropping the subscript 1). Then

$$\sqrt{n}\left(\frac{s^2}{\sigma^2}-1\right) = \sqrt{n}\left(\frac{n}{n-1}(m_2-\bar{z}^2)-1\right) = \sqrt{n}\,(m_2-1) - \sqrt{n}\,\bar{z}^2 + \frac{\sqrt{n}}{n-1}\,(m_2-\bar{z}^2)$$

$\sqrt{n}\,\bar{z}^2 = (\sqrt{n}\,\bar{z})\,\bar{z} \to 0$ as $\sqrt{n}\,\bar{z}$ is asymptotically $N(0,1)$ and $\bar{z} \to 0$

$\dfrac{\sqrt{n}}{n-1}(m_2-\bar{z}^2) \to 0\cdot(1-0) = 0$ as n increases

thus

$$\mathbb{E}\big(\sqrt{n}\,(m_2-1)\big) = 0, \qquad \mathrm{Var}\big(\sqrt{n}\,(m_2-1)\big) = \frac{1}{n}\,\mathrm{Var}\left(\sum z_i^2\right) = \mathrm{Var}(z^2) = \mathbb{E}z^4 - 1 = \kappa - 1$$

and by the CLT

$$\sqrt{n}\left(\frac{s_1^2}{\sigma^2}-1\right) \Rightarrow N(0, \kappa-1).$$

delta method with $g(\theta) = \log\theta$

$$\sqrt{n}\log\left(\frac{s_1^2}{\sigma^2}\right) \to N\left(0, \left(\frac{\partial g(\theta)}{\partial\theta}\right)^2(\kappa-1)\right), \qquad \left(\frac{\partial g(\theta)}{\partial\theta}\right)^2 = \left(\frac{1}{\theta}\right)^2 = \left(\frac{1}{1}\right)^2 = 1$$

$$\sqrt{n_1}\log\left(\frac{s_1^2}{\sigma^2}\right) \Rightarrow N(0,\kappa-1), \qquad \sqrt{n_2}\log\left(\frac{s_2^2}{\sigma^2}\right) \Rightarrow N(0,\kappa-1)$$

independent, so

$$\mathbb{E}(T) = \sqrt{\frac{n_2}{n_1+n_2}}\cdot 0 - \sqrt{\frac{n_1}{n_1+n_2}}\cdot 0 = 0$$

$$\mathrm{Var}(T) = \left(\frac{n_2}{n_1+n_2} + \frac{n_1}{n_1+n_2}\right)(\kappa-1) = \kappa-1$$

and $T \Rightarrow N(0,\kappa-1)$.

(d)

$$P(T > 1.645\sqrt{2}) = P\left(Z > \frac{1.645\cdot\sqrt{2}}{2}\right) = 0.122$$

we have a 12.2% chance of rejecting the null if $\kappa = 5$ is the true distribution.

(e)

> p-value robust to non-normality $= 0.175$, $\quad$ p-value under normality $= 0.047$

The p-value under normal assumptions shows that we would reject the null with 0.05 significance level, while the non-normality robust test shows we would not reject the null.

Problem 4

(a) $Y_i = \beta_j + \epsilon_i$ if alien i is of type j, so $x_i = j_i$ is the vector of type indicators and

$$\sqrt{n}(\hat{\beta}-\beta) \xrightarrow{d} N\left(0, \sigma^2(\mathbb{E}(xx'))^{-1}\right)$$

$\sigma^2 = \mathrm{Var}(\epsilon_i \mid j_i) = 1$ stated in problem.

j is indicator which has $\mathbb{E}(I^2) = \mathbb{E}(I) = 1/p$, so $\mathbb{E}(j_i j_i') = \frac{1}{p} I_p$ and

$$\mathbb{E}(xx')^{-1} = \mathbb{E}(j_i j_i')^{-1} = p\cdot I_p$$

$$\sqrt{n}(\hat{\beta}-\beta) \xrightarrow{d} N(0,\ p\,I_p), \qquad \text{each component } N(0,p).$$

(equivalently $\hat{\beta}_j$ is the mean of the n_j aliens of type j, with $n_j/n \to 1/p$.)

(b)

$$\min\ \|Y - Xb\|^2 + \lambda\sum|b_j|$$

$$\|Y-Xb\|^2 = \sum_j^p\sum_i^{n_j}(Y_i - b_j)^2 = \sum_j^p\sum_i^{n_j}(Y_i-\hat{\beta}_j)^2 + \sum_j^p n_j(\hat{\beta}_j - b_j)^2 = \sum_j^p n_j(\hat{\beta}_j-b_j)^2 + \sum\epsilon'$$

($\sum \epsilon'$ can't minimize with b_j)

$$\min \sum_j^p n_j(\hat{\beta}_j - b_j)^2 + \lambda \sum |b_j|$$

which separates across j. FOC $b_j \neq 0$: $\quad -2n_j(\hat{\beta}_j - b_j) + \lambda \operatorname{sign}(b_j) = 0 \qquad 2n_j\hat{\beta}_j = 2n_j b_j + \lambda \operatorname{sign}(b_j)$

$$b_j = \hat{\beta}_j - \operatorname{sign}(b_j)\frac{\lambda}{2n_j}$$

$$\hat{\beta}_{j,Lasso} = \begin{cases} \hat{\beta}_j + \dfrac{\lambda}{2n_j} & \hat{\beta}_j < -\dfrac{\lambda}{2n_j} \\ \hat{\beta}_j - \dfrac{\lambda}{2n_j} & \hat{\beta}_j > \dfrac{\lambda}{2n_j} \\ 0 & \text{otherwise} \end{cases}$$

(c) Take $\beta_j > 0$ (then $\hat{\beta}_j > \lambda/2n_j$ with probability $\to 1$ below, so the middle case of (b) applies). Here $n_j/n \to 1/p$, i.e. $\dfrac{n}{n_j} \to p$.

$\lambda = \sqrt{n}$

$$\sqrt{n}(\hat{\beta}_{j,Lasso} - \beta_j) = \sqrt{n}(\hat{\beta}_j - \beta_j) - \sqrt{n} \cdot \frac{\sqrt{n}}{2n_j} = \sqrt{n}(\hat{\beta}_j - \beta_j) - \frac{n}{2n_j}$$

the second term is non-random in the limit, $\dfrac{n}{2n_j} \to \dfrac{p}{2}$, so it shifts the mean and leaves the variance

$$\operatorname{Var}\left(\sqrt{n}(\hat{\beta}_{j,Lasso} - \beta_j)\right) \to \operatorname{Var}\left(\sqrt{n}(\hat{\beta}_j - \beta_j)\right) = p$$

$$\sqrt{n}(\hat{\beta}_{j,Lasso} - \beta_j) \xrightarrow{d} N\Big(-\frac{p}{2},\, p\Big)$$

$\lambda = n^{1/4}$

Var is unchanged but expectation will change

$$-\sqrt{n} \cdot \frac{n^{1/4}}{2n_j} = -\frac{n^{3/4}}{2n_j} = -\frac{p}{2} \cdot \frac{1}{n^{1/4}} \to 0$$

$$\sqrt{n}(\hat{\beta}_{j,Lasso} - \beta_j) \xrightarrow{d} N(0, p)$$

$\lambda = n^{3/4}$

$$-\sqrt{n} \cdot \frac{n^{3/4}}{2n_j} = -\frac{n^{5/4}}{2n_j} = -n^{1/4} \cdot \frac{p}{2} \to -\infty$$

so $\sqrt{n}(\hat{\beta}_{j,Lasso} - \beta_j)$ has variance p but its mean diverges to $-\infty$: there is no limiting distribution (estimator is biased by too much).

(d) For a useless type $\beta_j = 0$, $\hat{\beta}_{j,Lasso} = 0$ when $|\hat{\beta}_j| \leq \left|\dfrac{\lambda}{2n_j}\right| = \dfrac{\sqrt{n}}{2n_j}$, and multiplying by $\sqrt{n}$ the bound is $\dfrac{n}{2n_j} \to \dfrac{p}{2}$

$$P\Big(-\frac{\sqrt{n}}{2n_j} \leq \hat{\beta}_j \leq \frac{\sqrt{n}}{2n_j}\Big) = P\Big(-\frac{n}{2n_j} \leq \sqrt{n}\hat{\beta}_j \leq \frac{n}{2n_j}\Big) \to P\Big(-\frac{p}{2} \leq \sqrt{n}\hat{\beta}_j \leq \frac{p}{2}\Big)$$

$\sqrt{n}(\hat{\beta}_j - \beta_j) = \sqrt{n}\hat{\beta}_j \to N(0, p)$ as $\beta_j = 0$ if useless alien, so

$$P\left(-\frac{p}{2} \leq \sqrt{n}\hat{\beta}_j \leq \frac{p}{2}\right) = 2\Phi\left(\frac{p}{2} \cdot \frac{1}{\sqrt{p}}\right) - 1 = 2\Phi\left(\frac{\sqrt{p}}{2}\right) - 1$$

Φ is normal cdf. The $\hat{\beta}_j$ are asymptotically independent (diagonal covariance), so for having k useless aliens we need all k of them to be zero, $\left(2\Phi\left(\frac{\sqrt{p}}{2}\right) - 1\right)^k$, while types with $\beta_j > 0$ are nonzero with probability $\to 1$ (so only the useless ones are zero).

> asymptotic probability of k useless aliens is $\left(2\Phi\left(\frac{\sqrt{p}}{2}\right) - 1\right)^k$

Problem 5

No. Strict stationarity means the joint distribution of a sequence doesn't depend on time t. The sequence can have auto-correlation and heteroskedasticity in this case. The distribution of (y_t, x_t) is constant, but if we condition the distribution on (y_{t-1}, x_{t-1}) then the distribution on $y_t \mid x_t$ can vary. Stationarity only enforces that the variance-covariance matrix doesn't depend on t, but it can depend on $t - 1$ information if conditioned on it.